

SPSS first steps: describe variables, then describe apparent associations.

Theory deck for Worksheet 1: NHANES 1999-2003, descriptive statistics, crosstabs, risk measures, scatterplots, covariance, Pearson, and Spearman.

Worksheet objective: choose, compute, interpret

01 Inspect

Variable view, coding, units

02 Classify

Nominal, ordinal, metric

03 Plot

Match graph to variable type

04 Compare

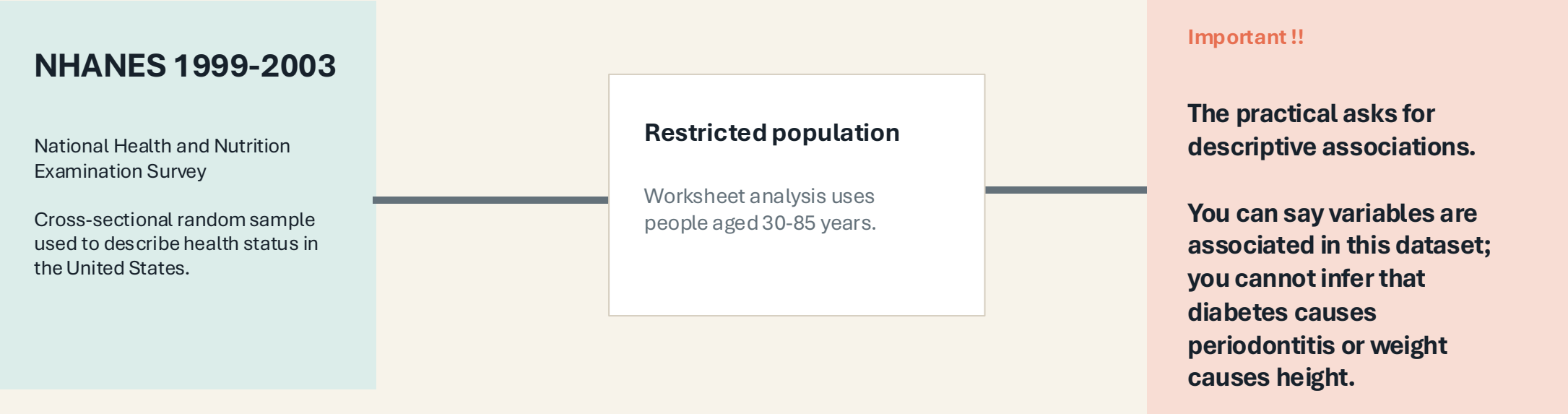
Use crosstabs or correlations

Core rule

The measurement level decides the graph, summary, and association measure.

In a cross-sectional study, describe associations carefully. Do not turn them into causal conclusions.

NHANES describes population health, so interpretation starts with study design.



Oral health

nteeth, ca_l_mean, perio, perio2, bop

Demographics

male, age, edu

Lifestyle

smk, smk2, alcg

Body measures

Height, Weight, bmi, wc

Health outcome

diab, mortstat, perm_exm

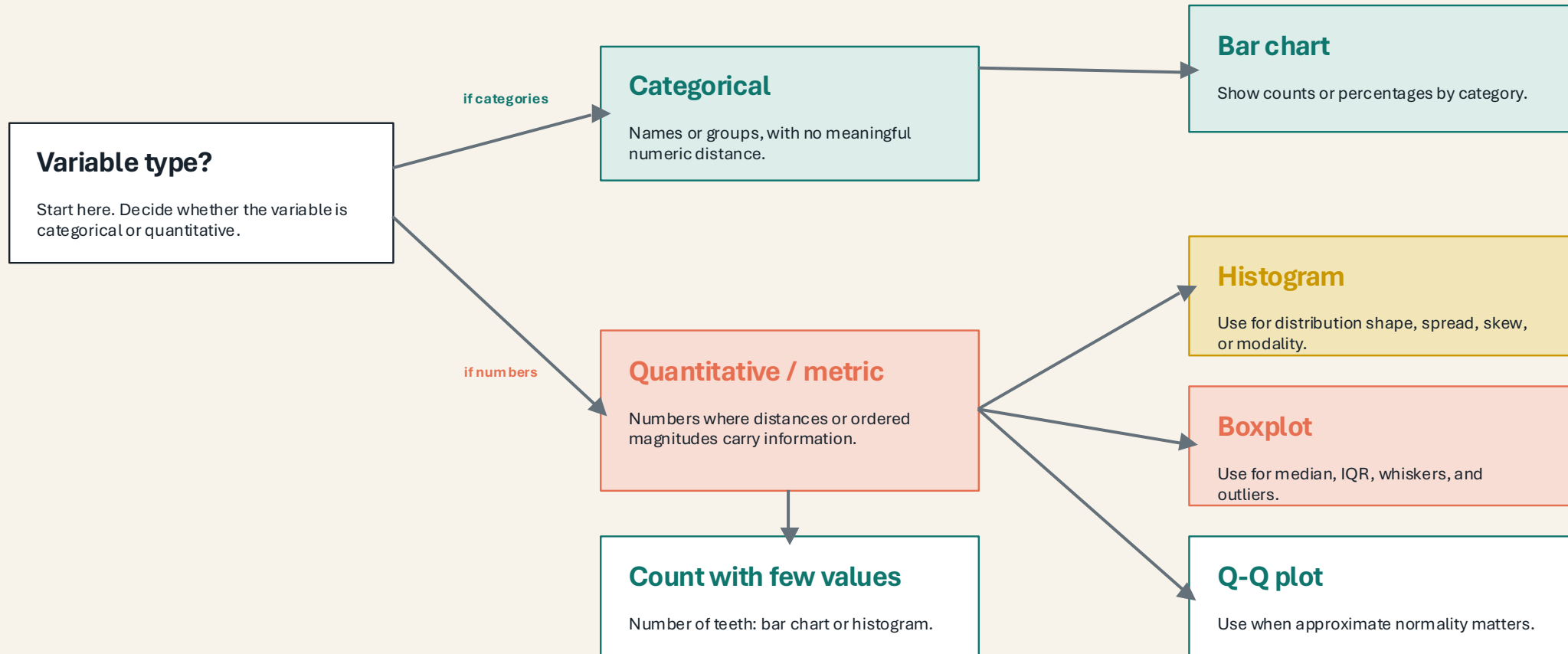
Measurement level determines what “summarize” and “compare”

Variable	Level	What values mean	Good summaries	Good graphics
Sex (male)	Nominal	Categories, no order	Counts, percentages	Bar chart
BMI category	Ordinal	Ordered groups	Counts, percentages, median if coded carefully	Bar chart
BMI	Metric, numeric, continuous	Numeric distance is meaningful	Mean, SD, median, IQR	Histogram, boxplot
Number of teeth	Discrete metric	Count from 0 to 32	Mean/median, distribution, floor effects	Bar or histogram
Height	Metric, numeric, continuous	Measured in cm	Mean, SD, range	Histogram, boxplot

SPSS check: open Variable View and confirm Measure is set consistently with the actual coding.

Key distinction: number of teeth is a count with bounded integer values; height and BMI are continuous measurements.

Choose the graph from the question and the variable type before using Chart Builder.

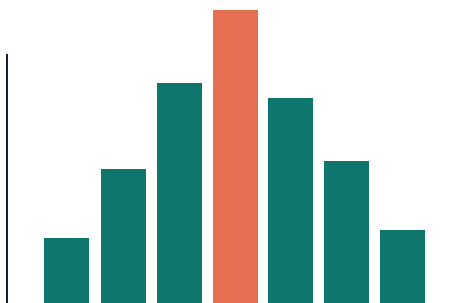


SPSS path: Graphs -> Chart Builder. Pick the chart family first, then drag the variable to the axis.

Distribution graphics answer different questions about the same metric variable.

Histogram

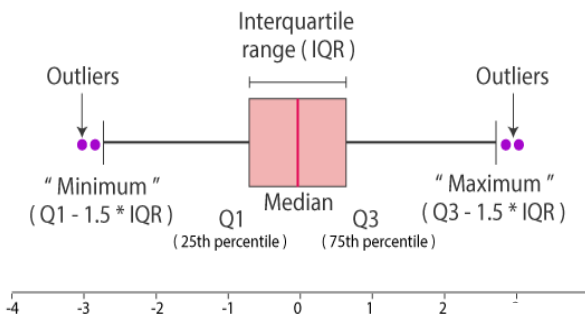
What is the shape?



Use for BMI, attachment loss, height, or weight when you need shape and spread.

Boxplot

Where are median, IQR, and outliers?

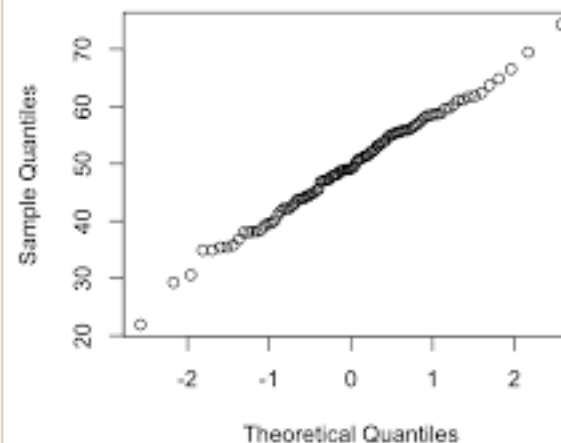


Different parts of boxplot © Byjus.com

Boxplots compress the distribution into median, quartiles, whiskers, and possible outliers.

Q-Q plot

Does the distribution roughly follow normality?



Interpret first, compute second: a plot can reveal coding errors, outliers, skewness, and impossible values before formal analysis.

A crosstab turns diabetes and periodontitis into conditional risks.

	Periodontitis yes	Periodontitis no	Row total
Diabetesyes	a	b	a + b
Diabetes no	c	d	c + d
Column total	a + c	b + d	n

SPSS path: Analyze -> Descriptive Statistics -> Crosstabs. In Cells, request row percentages. In Statistics, select Risk.

A useful convention is to put the outcome across the top and the exposure in rows.

Difference, ratio, and odds answer three different association questions.

	Periodontitis yes	Periodontitis no	Row total
Diabetesyes	a	b	a + b
Diabetesno	c	d	c + d
Column total	a + c	b + d	n

Risk difference

How many percentage points higher or lower?

$$RD = [a / (a+b)] - [c / (c+d)]$$

Use percentage points, not percent.

Relative risk

How many times as high is the risk?

$$RR = [a / (a+b)] / [c / (c+d)]$$

RR = 1 means no association.

Odds ratio

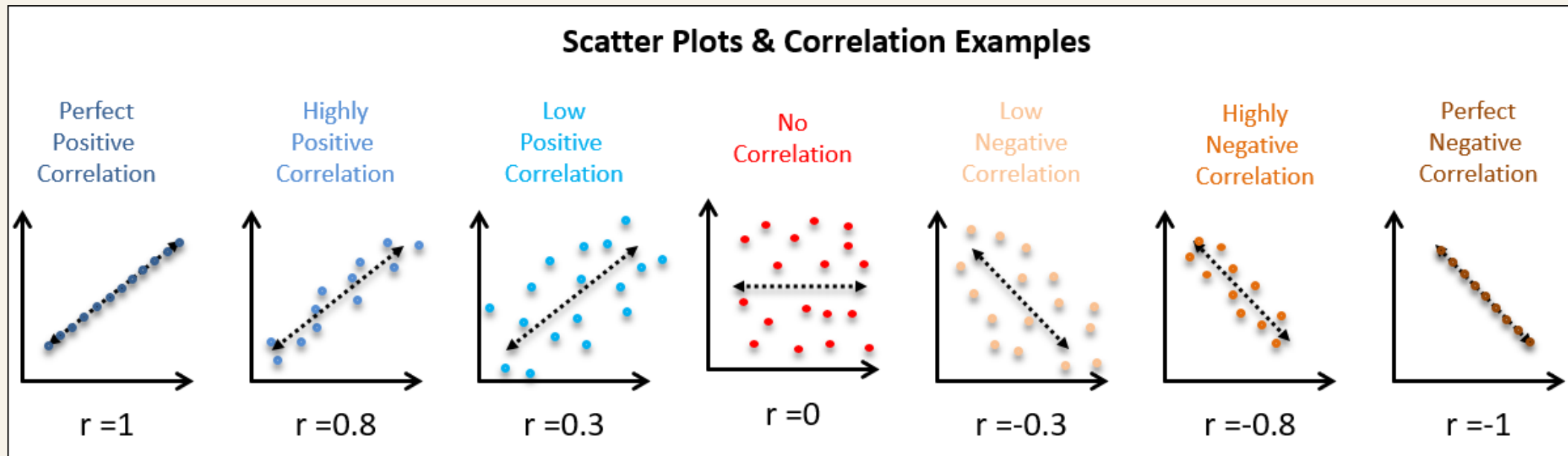
How many times as high are the odds?

$$OR = (a/b) / (c/d) = ad/bc$$

OR approximates RR only for rare outcomes.

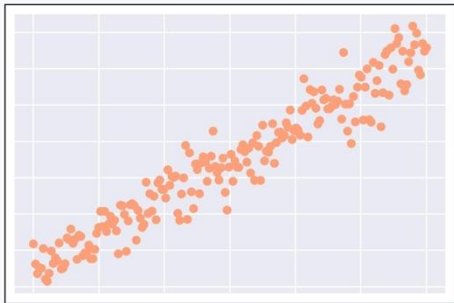
Hand-check SPSS: subtract row percentages for RD, divide row percentages for RR, and verify the odds ratio from the four table cells.

A scatterplot shows the relationship before a coefficient compresses it.



Pearson vs Spearman correlation.

Linear Data

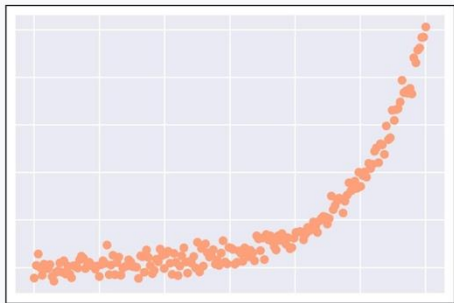


Pearson
Correlation **0.96**

Spearman
Correlation **0.96**

Same

Non-linear Data



Pearson
Correlation **0.76 X**

Spearman
Correlation **0.92 ✓**

Use **Pearson correlation** when your goal is to measure a linear relationship between two continuous variables under fairly standard statistical assumptions:

- Variables are continuous.
- Relationship is roughly linear.
- The distribution is not strongly skewed and doesn't include extreme outliers.

The **Spearman's** test can be used to analyse ordinal level, as well as continuous level data, because it uses ranks instead of assumptions of normality.

Spearman correlation is helpful when the data or relationship does not fit Pearson's assumptions:

- Ordinal or rank-based data.
- Monotonic relationships that may not be linear.
- Non-normal or skewed distributions or outliers sensitive

